C29xMCAL_01.04.00 Release Notes

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Introduction

F29x MCAL is a cohesive set of tools for AUTOSAR based development on C2000 real-time controllers. It includes device-specific drivers (MCALs), EB Tresos Plugins for configuration, Bare-metal Examples for each driver.

F29H85x MCAL v01.04.00 is an early adopter release targeted for F29H85x.

Device Support

The following devices are supported with this release

- MCU's
  - F29H859DU_Q1
  - F29H859TM_Q1
  - F29H859TU_Q1
  - F29P329SJ_Q1
  - F29P329SM_Q1
  - F29P589DM_Q1
  - F29P589DU_Q1
- Hardware Platforms
  - F29H85X-SOM-EVM

What Is Supported

- MCAL / CDD (AUTOSAR **v4.3.1**):

Drivers	**Modules**
Micro controller Drivers	GPT, MCU, WDG
Communication Driver	CAN, LIN, SPI
I/O Drivers	DIO, PORT
Memory Drivers	FLS
Complex Device Drivers	ADC, XBAR, SENT, IPC, UART, PWM, ECAP, I2C

- Resource Allocator
- Reference Code:
  - Multi-Core startup code
- Build System:
  - CMake
  - CCS Projectspec

Note: The minimum version of CCS Theia required is CCSv20.03.00 or above and compiler version is C29Clang 2.0.0 STS. Please refer to the Dependencies section for the download links.

What Is Not Supported

- Multi-core MCALs as per AUTOSAR 4.3.1.

New In This Release

- MCAL / CDD :
  - Existing modules with bug fixes -
    1. MCU
    2. GPT
    3. OS(Stubs)
    4. CAN
    5. CDD ECAP
    6. CDD PWM
    7. FLS
    8. CDD UART
    9. CDD SENT
    10. LIN
    11. PORT
  - New driver added -
    1. CDD I2C
  - New Functionality added -
    1. PORT
       1. Resource allocator is integrated
    2. CAN
       1. Resource allocator is integrated
    3. LIN
       1. Resource allocator is integrated
    4. SPI
       1. Resource allocator is integrated
    5. FLS
       1. Resource allocator is integrated
    6. CDD ADC
       1. Resource allocator is integrated
       2. New example for SDK DMA integration with CDD ADC is added
       3. New API is added: Cdd_Adc_UpdateStatusThroughDma
    7. CDD XBAR
       1. Resource allocator is integrated
       2. New example for external interrupt is added
    8. CDD SENT
       1. Resource allocator is integrated
    9. CDD PWM
       1. Resource allocator is integrated
       2. New example for SDK CMPSS integration with CDD PWM is added
    10. CDD ECAP
       1. New API is added: Cdd_Ecap_PeriodicReadback
    11. CDD IPC
       1. Resource allocator is integrated

12. CDD UART
    1. Resource allocator is integrated.
    2. New API is added: Cdd_Uart_GetErrorStatus
    3. New API is added: Cdd_Uart_FlushReadFIFO
13. Mcal_Lib
    1. Memmap area is added
14. MCU
    1. New memory sections are added
    2. Exclusive area is added
    3. New API is added: Mcu_getSystemClock
15. hw_includes
    1. All hw registers related files are updated as per latest specification

- Support added for F29P32x and F29P58x device variants
  - Added examples for F29P32x and F29P58x devices
  - Reference linker files are updated
  - Target configs are added for F29P32x and F29P58x devices
- Build Infrastructure
  - BUILD_DEVICE_F29H85x/ BUILD_DEVICE_F29P58X/ BUILD_DEVICE_F29P32X macro can be used to build device specific examples. It will emulate the F29H85x hardware as necessary device.

Module Version Number

MCU	02.01.00
PORT	03.00.00
DIO	01.00.03
GPT	02.00.01
CAN	03.00.00
WDG	02.00.00
LIN	03.00.00
SPI	03.00.00
FLS	03.00.00
CDD ADC	03.00.00
CDD XBAR	03.00.00
CDD SENT	03.00.00

CDD PWM	03.00.00
CDD ECAP	03.00.00
CDD IPC	02.00.00
CDD UART	03.00.00
CDD I2C	01.00.00

Note:

We follow Autosar's requirement [SRS_BSW_00321] for versioning of each module-

5.2.4.10 [SRS_BSW_00321] The version numbers of AUTOSAR Basic Software Modules shall be enumerated according specific rules

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Type:	Valid
Description:	The version numbers of AUTOSAR Basic Software Modules shall be enumerated according to the following rules: - Increasing a more significant digit of a version number resets all less significant digits - The PATCH_VERSION is incremented if the module is still upwards and downwards compatible (e.g. bug fixed) - The MINOR_VERSION is incremented if the module is still downwards compatible (e.g. validfunctionality added) - The MAJOR_VERSION is incremented if the module is not compatible any more (e.g. existing API valid)]
Rationale:	Provide unambiguous version identification for each module, provide version cross check as well as basic version retrieval facilities. Compatibility is always visible!
Use Case:	Example: ADC module with version 1.14.2:

Fixed In This Release

Key	Summary	Component/s	Severity	P
MCAL-35093	Literals and Tags missing in Arxml file	All	S3-Minor	○
MCAL-35092	CAN: CanIf_TriggerTransmit should be called only if enabled	Can	S3-Minor	⌃⌃
MCAL-35091	CAN: Correct Multiplicity for CanMainFunctionRWPeriodRef and CanOsCounterRef as per AUTOSAR	Can	S3-Minor	○
MCAL-35062	CAN: Failure to transmit message after bus off recovery	Can	S3-Minor	⌃⌃
MCAL-35045	Cdd_Xbar_Init should be enabled even when CddXbarAPIEnable is false	CDD XBAR	S2-Major	⌃⌃
MCAL-35044	Error check if PortNumberOfPortPins doesn't match the number of pins is not working	Port	S2-Major	⌃⌃
MCAL-34910	Error is including Det.h in SPI driver.	Spi	S3-Minor	⌃⌃
MCAL-34842	API-SERVICE-PREFIX tag is missing in Cdd_Adc plugin	CDD ADC	S3-Minor	⌃⌃
MCAL-32743	EPWM Channel name not populating in output (.h)	CDD PWM	S3-Minor	⌃⌃
MCAL-32459	float32 and float64 defined in both hw_types.h and Platform_types.h	All	S3-Minor	○
MCAL-32433	SpiCsPolarity configuration should be the active state of the chip select	Spi	S3-Minor	⌃⌃
MCAL-32322	CAN: Can communication with 5Mbps data rate is not verified	Can	S3-Minor	○
MCAL-31535	Not able to send LIN frames after calling Lin_Wakeup()	Lin	S2-Major	⌃⌃
MCAL-31442	Incorrect uniqueness check for User Callback function in CDD SENT	CDD SENT	S3-Minor	○
MCAL-30754	Cdd Sent: Regression testing is not performed for 01.03.00 release	CDD SENT	S2-Major	⌃⌃

MCAL-30631	PortPinId validation code in incorrect in PORT module	Port	S2-Major	

16 issues

Known Issues And Limitations

There are few limitations and known issues associated with this release

Limitations

- CDD ECAP -
  - In case of high frequency signals (i.e. 500Khz and above), software overhead in user callback shall be kept to minimum to avoid missing edges.
- Test Infrastructure : Following tests have not been performed due to limitations of available test infrastructure -
  - CDD SENT : Following modes in SENT are not tested -
    - Enhanced 16-bit Slow Mode.
    - Nibble Count 1,2,3,4,5.
    - MTP Mode with all 4 sensors connected.
  - CDD IPC : IPC is validated between CPU1 and CPU3. Since, CPU2 is configured in lockstep in our framework; IPC between CPU1 and CPU2 or CPU2 and CPU3 is not validated.
  - Fls : Timeout tests are not performed as FLASH characterization data is not yet available for this device.
  - Spi : Board to board Spi communication validated upto 40MHz. 50MHz have been only test in loopback mode.
  - CDD I2C: Target mode is not supported.
- Compiler
  - Link Time optimization is currently not supported.

New Known Issues

Key	Summary	Component/s	Workaround	Severity
MCAL-35347	I2C: Cancel API is not functional	CDD I2C	Do not cancel the ongoing transmission using API. Instead handle this situation in application code.	S2-Major

1 issue

Existing Known Issues

Key	Summary	Component/s	Workaround	Severity
MCAL-32321	All Modules: Link time optimization is not verified for all modules	All	By default LTO is disabled, user shall not enable it.	S2-Major

1 issue

Quality Status

Sr. No.	Work Product	Availability	Where to find Artifacts	Current Quality Status
1	Dynamic Analysis Report 1. MCDC 2. Branch 3. Region 4. Line 5. Function	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
2	Static Analysis Report 1. Static 2. HIS 3. MISRA-C	Yes	Installer:*\docs\Static_Analysis	All Mandatory & Critical warnings and errors are resolved. Required and Advisory warnings will be fixed in future releases.
3	Test Reports: Test Cases and Test Results	Yes	Installer:*\docs\User_Guide	All test reports are present in installer.
4	Bi-directional Traceability Report 1. MRD↔SPS 2. SPS↔SW Arch ↔SW Design Doc↔SW Units (Code) 3. MRD↔SPS↔Test Case 4. Test Case↔Arch Test Case↔Design	No	CSP	Will be available in future releases. Please refer to device roadmap for details.

5	Customer Docs: User Guide	Yes	Installer:*\docs\Test_Report	User Guide is present in installer as well as softwaredl
6	SW FMEA	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
7	Detailed Design Document	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
8	SW Manifest	Yes	Installer:*\mcal_manifest	SW manifest is present in installer
9	Software Product Specification: List of functional, safety requirements for SW components	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
10	Architecture Document	No	CSP	Will be available in future releases. Please refer to device roadmap for details.

11	SW Safety Manual	No	CSP	Will be available in future releases. Please refer to device roadmap for details.
12	Safety Assessment	No	CSP	Will be available in future releases. Please refer to device roadmap for details.

Dependencies

This release has dependency on the following tools

- CCS v20.03.00 or above version.
- C29Clang Compiler 2.0.0 STS.
- EB Tresos (ACG 8.8.12) : Request access here.

References

Technical Support

https://e2e.ti.com/support/microcontrollers/c2000/f/171